PHY102 Electricity and Magnetism Jan-April 2026: Assignment 7

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To be discussed in the Tutorial on March 27, 2026

1. Consider a circular path of radius r in the $x-y$ plane. Consider that there is a current I_1 flowing along the z axis and another current I_2 following along a line in $x-z$ plane passing through the origin and at an angle 45^0 with the z axis. Explicitly calculate the line integral of the magnetic field along the closed circular path defined above. The field is caused by both the currents.

2. Consider a charge q at origin and at rest at time $t=0$ that gains a velocity $\vec{v} = \frac{c}{10}\hat{x}$(i.e. $(\beta = 0.1)$ in a short time Δt and moves at this velocity upto $t = 10sec$ at this point of time the charge stops in a very short time Δt and remains stationary thereafter. Make sketches of the electric field at three times $t = 0sec$, $t = 5sec$ and $t = 15sec$

3. Show that Divergence of Curl of a vector field is always zero.
$$\nabla.(\nabla \times \vec{F}) = 0$$

4. Imagine a current density given by $\vec{J} = J_0\hat{x}$. Consider a circular disc of radius r at distance d from the origin with its center on the x axis and lying in the $y-z$ plane. Calculate the current flowing through this disc. Now imagine a hemispherical cap on this disc, explicitly calculate the current through this hemispherical cap. Interpret your answer.